

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9744/02

Paper 2 Structured Questions

25 August 2017

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and class in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
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This document consists of **26** printed pages and **4** blank pages.

[Turn over

Answer **all** questions.

- 1 The plasma membrane Ca^{2+} ATPase (PMCA) is a vital transport protein that regulates the amount of Ca^{2+} within eukaryotic cells. In humans, there is a very large transmembrane electrochemical gradient of Ca^{2+} driving the entry of Ca^{2+} into cells, yet low intracellular concentrations of Ca^{2+} are maintained by PMCA.

Fig. 1.1 shows two conformations that PMCA interconverts between, depending on whether Ca^{2+} is bound.

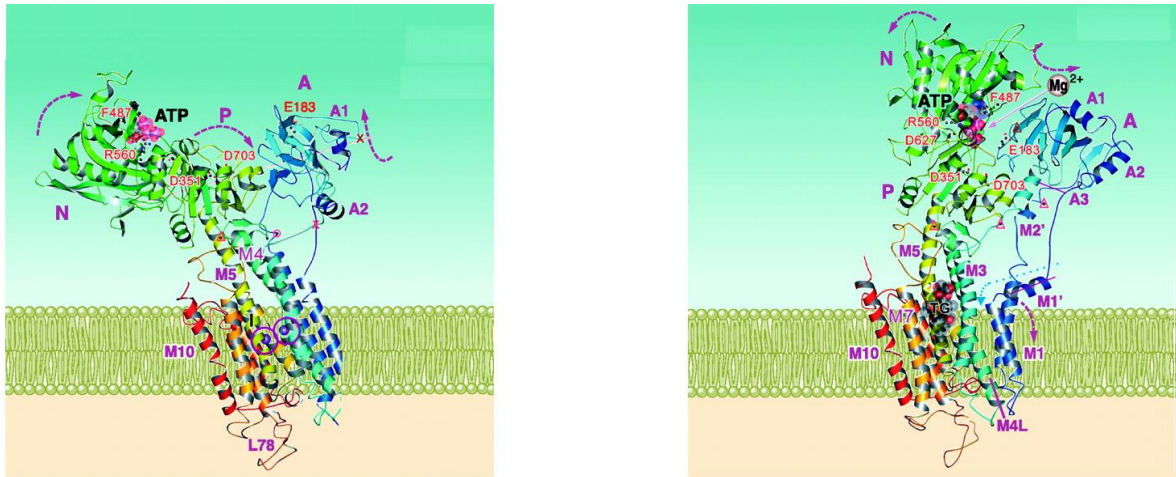


Fig. 1.1

- (a) Explain why Ca^{2+} cannot freely cross the plasma membrane. [2]

- (b) Describe how low intracellular concentrations of Ca^{2+} are maintained by PMCA. [3]

Various forms of PMCA are expressed in different cell types, including erythrocytes (red blood cells). Erythrocytes rely on Ca^{2+} dependent signalling during their differentiation from hematopoietic stem cells in the bone marrow. The process of erythropoiesis (production of mature red blood cells) is illustrated in Fig. 1.2.

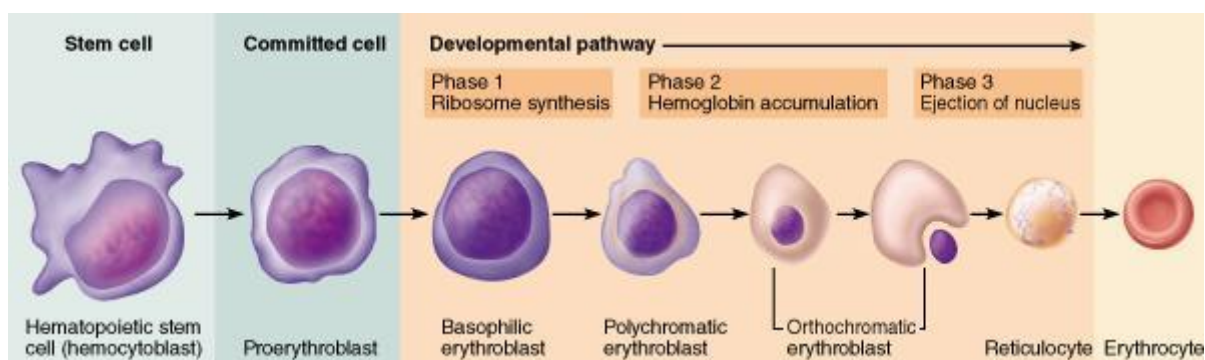


Fig. 1.2

(c) Describe the potency of these hematopoietic stem cells and explain their normal functions. [4]

In addition to the loss of the nucleus, cellular organelles such as the endoplasmic reticulum, Golgi apparatus and mitochondria are also lost from the reticulocytes shown in Fig. 1.2.

(d) Outline the structure of the endoplasmic reticulum and Golgi apparatus in typical eukaryotic cells. [2]

- (e) Since mature mammalian erythrocytes lack mitochondria, suggest how these cells derive energy from glucose. [1]

[Total: 12]

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2 Fig. 2.1 shows two animal cells in different stages of the mitotic cell cycle.

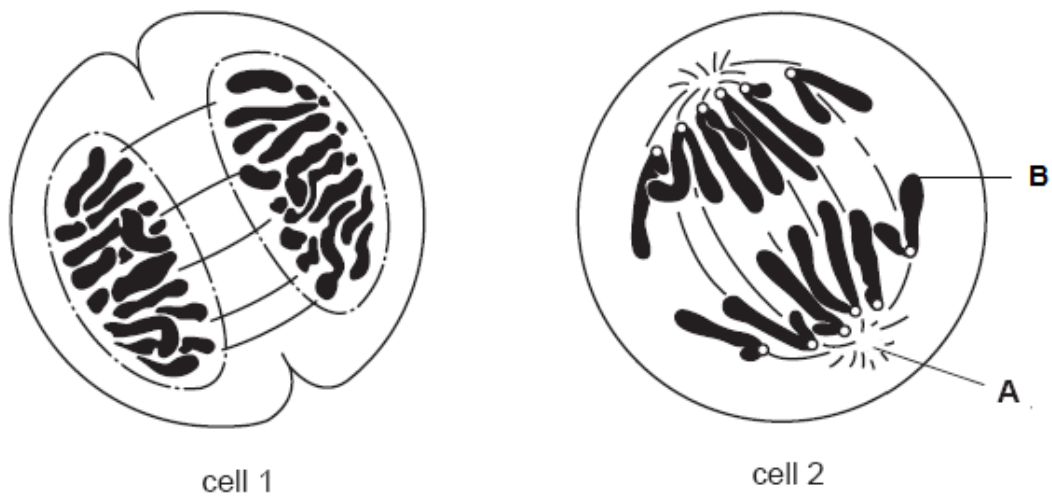


Fig. 2.1

(a) With reference to cell 1,

(i) identify the stage of nuclear division taking place. [1]

(ii) describe the events occurring at this stage. [2]

(b) A pair of rod-like structures can be found in region A. Outline the roles of these structures during mitosis in animal cells. [2]

(c) Region B of the chromatid contains non-coding repetitive nucleotide sequences.

(i) Account for the progressive shortening of these sequences after repeated rounds of DNA replication. [2]

(ii) Describe two functions of these non-coding DNA in eukaryotes. [2]

- (d) In 2006, Yamanaka and his colleagues demonstrated in an experiment with mice that induced pluripotent stem (iPS) cells could be produced by genetically reprogramming fully differentiated adult cells. There has been evidence to suggest that these iPS cells exhibit high telomerase reverse transcriptase (TERT) activity and are capable of dividing indefinitely.

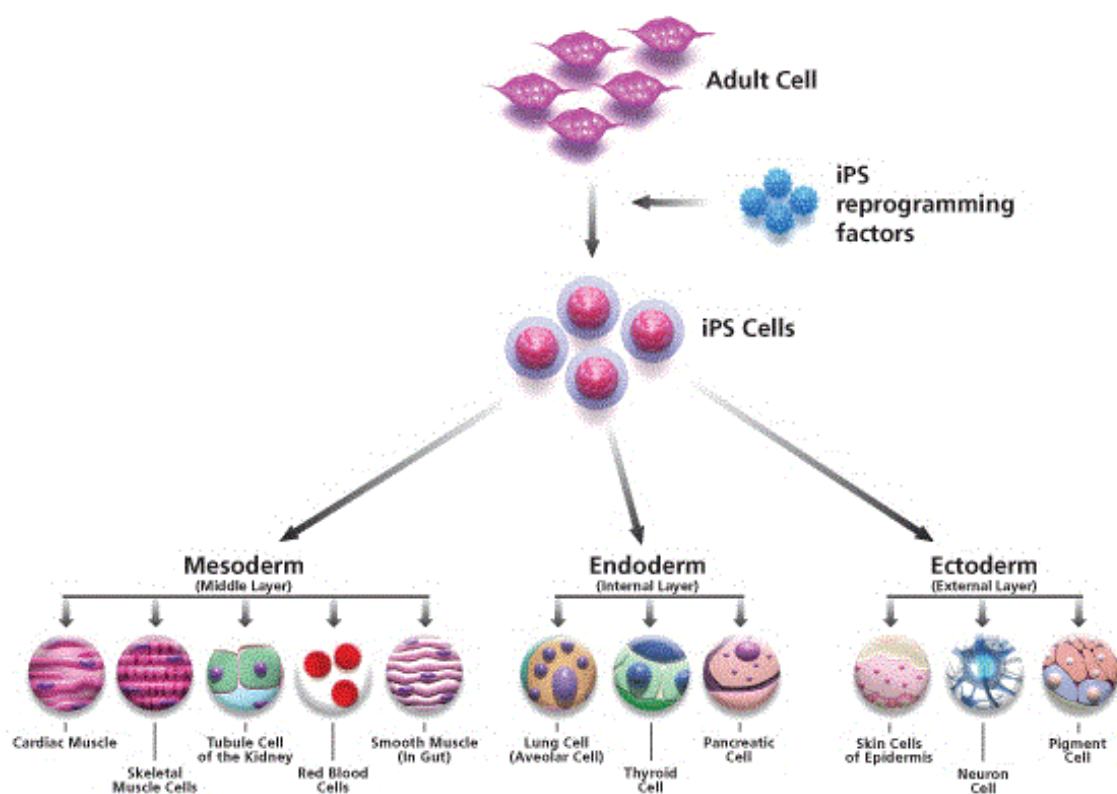


Fig. 2.2

- (i) Discuss the role of TERT in enabling the iPS cells to divide indefinitely. [2]

- (ii) Suggest an advantage of using iPS cells in research and medical applications. [1]

[Total: 12]

- 3** Antibiotic resistance is rising to dangerously high levels in all parts of the world. A growing list of infections – such as pneumonia, tuberculosis, blood poisoning and gonorrhoea – are becoming harder, and sometimes impossible, to treat as antibiotics become less effective.

Some bacteria are naturally resistant to certain types of antibiotics. However, bacteria may also become resistant either by a genetic mutation or by acquiring resistance from another bacterium.

- (a)** Outline the process of how a bacterium is able to acquire resistance from another bacterium. [3]

More than 2 million Americans each year are infected by antibiotic-resistant bacteria, and at least 23,000 die annually from those infections. Antibiotic-resistant bacteria have become a global health crisis and alternative treatments such as Phage Therapy are being considered for combating bacterial infections.

Phage Therapy involves the targeted application of bacteriophages that, upon encounter with specific pathogenic bacteria, can infect and kill them. Phages are currently being used therapeutically to treat bacterial infections that do not respond to conventional antibiotics.

Fig. 3.1 is an electron micrograph showing a phage infecting a bacterium during Phage Therapy.

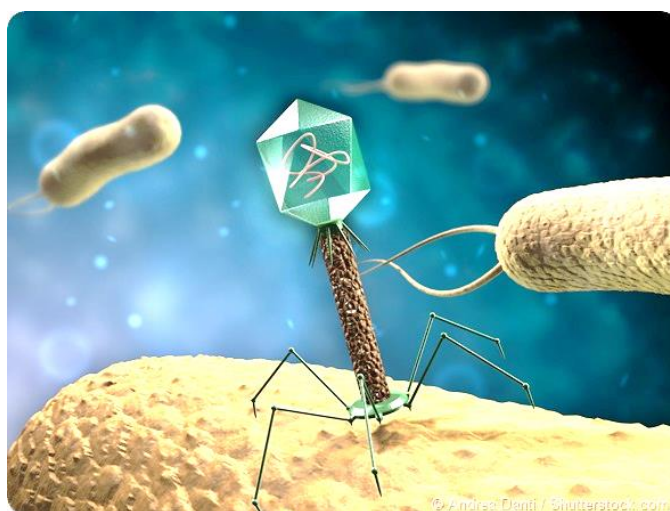


Fig. 3.1

(b) Suggest the reproductive cycle of a phage used for Phage Therapy. [1]

(c) Describe how a structural feature of the phage allows for targeted application to specific pathogenic bacteria. [2]

(d) Explain how the use of phages can prevent the spread of bacterial infection. [2]

(e) Suggest characteristics of phages that make them attractive therapeutic agents. [2]

[Total: 10]

- 4 Sickle cell anaemia is a recessive genetic disease caused by a mutation that commonly occurs in the DNA, resulting in hydrophobic valine replacing hydrophilic glutamic acid at the 6th amino acid position of the β chain.

(a) State the type of mutation that commonly occurs to result in sickle cell anaemia. [1]

(b) Describe the effects of this change in amino acids on the red blood cells of an individual with the disease. [4]

To detect if individuals are afflicted with sickle cell anaemia, restriction fragment length polymorphism (RFLP) analysis can be carried out using gel electrophoresis and Southern Blotting. Restriction enzymes are used to digest the DNA before RFLP analysis and the mutation removes a recognition site of the restriction enzyme *MstII*, as shown in Fig. 4.1. The enzyme's recognition sites on the normal allele and the mutant allele are shown by arrows.

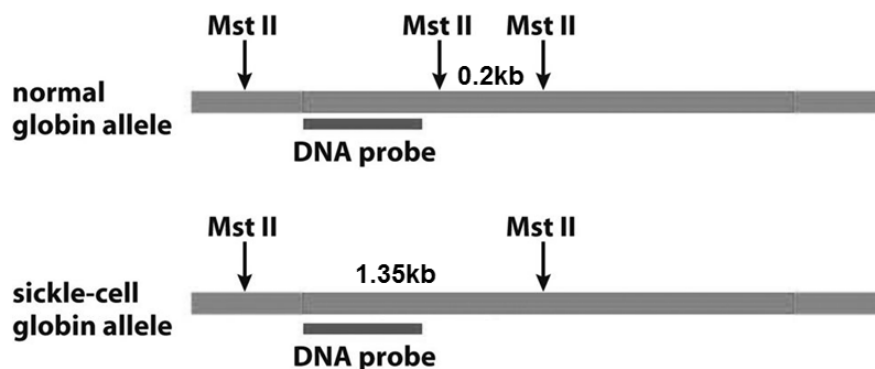


Fig. 4.1

- (c) Draw, on Fig. 4.2, the expected band patterns produced by DNA from individuals with sickle cell anaemia. [1]

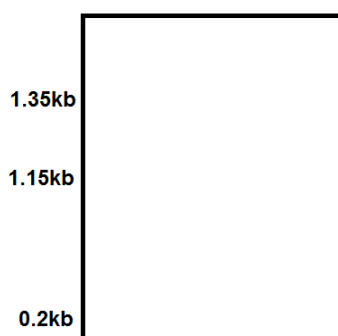


Fig. 4.2

- (d) Suggest why it is necessary to carry out Southern Blotting after gel electrophoresis. [2]

- (e) Outline the role of the DNA probe in Southern Blotting. [1]

[Total: 9]

- 5 Fig. 5.1 shows awned and awnless rice strains. A long awn is one of the distinct morphological features of wild rice species. It is a long needle-like appendage that is thought to aid in seed dispersal and prevent predation by animals.

The genes *DROOPING LEAF (DL)* and *OsETTIN2 (OsETT2)* are involved in awn formation. Genetic analysis experiments indicate that *DL* and *OsETT2* act independently in awn formation.



Fig. 5.1

A cross between pure-breeding awned and awnless strains produced awned plants in F1.

The F1 plants were then self-pollinated.

In the F2 generation, 658 awned plants and 48 awnless plants were produced.

This control of awn development is an example of epistasis resulting in a ratio that is close to 15:1.

(a) Define the term *locus*. [2]

(b) Explain the term epistasis in this context. [3]

- (c) Use the symbols A, a and B, b to draw a genetic diagram to explain the results shown in the F₂ generation. [4]

A chi-squared test was carried out on the results of the second cross.

Table 5.1

phenotype	observed number (O)	expected number (E)	$\frac{(O - E)^2}{E}$
awn	658		
awnless	48		
total number	706	706	$\chi^2 =$

(d) Complete the five missing values in Table 5.1. [3]

(e) Table 5.2 shows part of the table of probabilities for the chi-squared test.

Table 5.2

degrees of freedom	probability								
	0.995	0.975	0.9	0.5	0.1	0.05	0.025	0.01	0.005
1	.000	.000	0.016	0.455	2.706	3.841	5.024	6.635	7.879

Use Table 5.2 and your calculated value for the chi-squared test to find the probability that the observed ratio of phenotype does not deviate significantly from the expected ratio. [1]

(f) State what conclusions may be drawn from the probability found in (e). [2]

[Total: 15]

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6 Fig. 6.1 shows some stages in mammalian respiration.

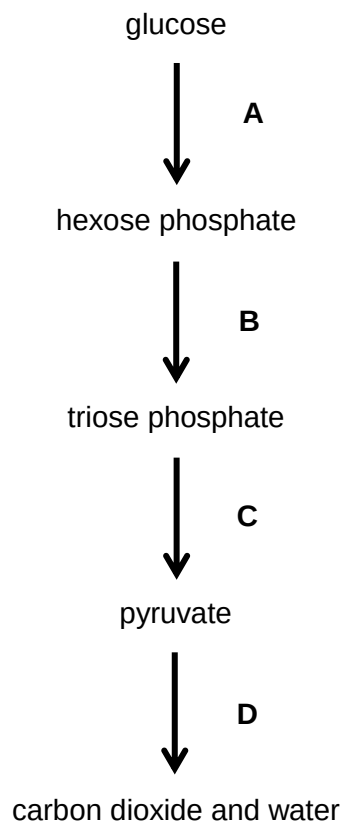


Fig. 6.1

(a) Name the processes taking place during Stage D and state precisely where they occur. [3]

(b) Intermediates produced at the end of Stages B and C are important in the conversion of carbohydrates to lipids such as triglycerides. Some of the triose phosphate can be converted into glycerol-3-phosphate, while pyruvate can undergo further reactions to form intermediates required for the synthesis of fatty acids.

(i) Describe the formation of triglycerides. [3]

(ii) State two roles of triglycerides in living organisms. [2]

- (c) The first reaction in Stage A is catalysed by the enzyme hexokinase. It has been observed that hexokinase is bound to the outer mitochondrial membrane in muscle cells which undergo high rates of glycolysis.

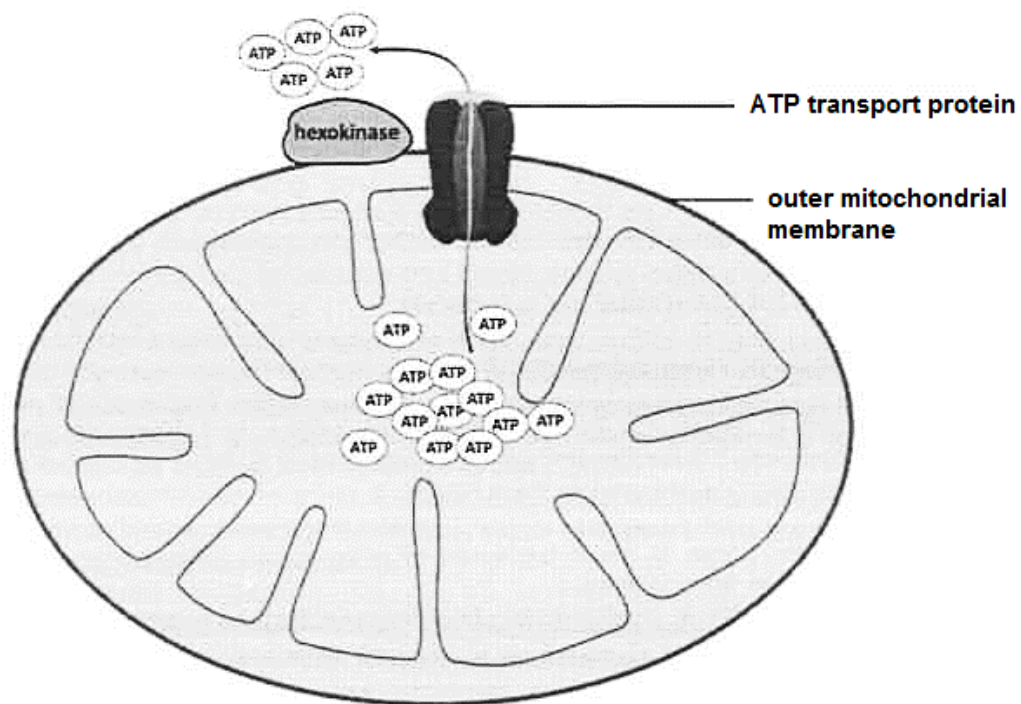


Fig. 6.2

With reference to the role of mitochondria and Fig. 6.2, suggest how the association of hexokinase with mitochondria can lead to high rates of glycolysis. [2]

Fig. 6.3 shows an electron micrograph of a mitochondrion.

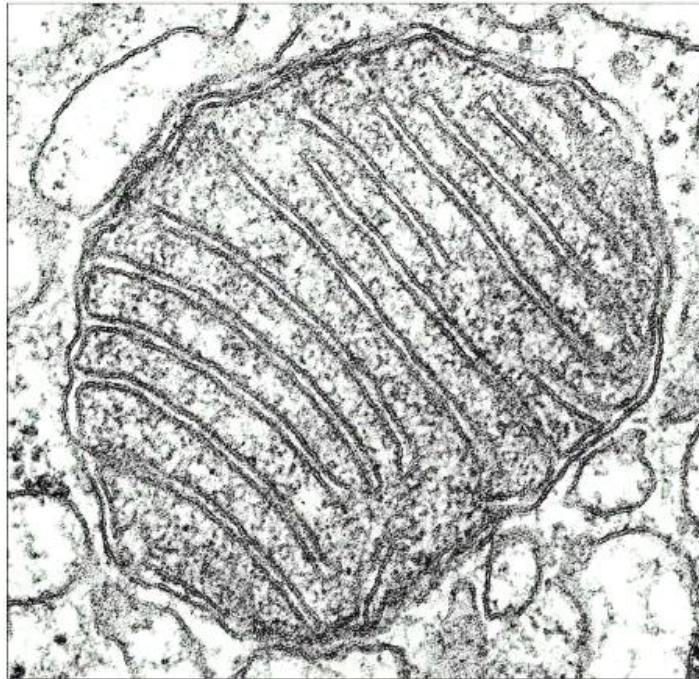


Fig. 6.3

(d) With reference to features visible in Fig. 6.3, outline how the structure of the mitochondrion is adapted for its function. [2]

[Total: 12]

- 7 Fig. 7.1 shows a flower of *Lilium polyphyllum*, a lily that grows in the Himalayan mountains. This species is cross-pollinated by insects.



Fig. 7.1

- (a)** Plants of this species that grow at low altitudes produce flowers 60 days before the plants of the same species that grow at high altitudes. Scientists think that plants of *L. polyphyllum* growing at high altitudes may evolve into a new species.

Explain how natural selection could lead to the evolution of a new species of lily. [5]

- (b) In order for natural selection to occur a population must show phenotypic variation. Explain why variation is important in natural selection. [2]

- (c) Fungi were often classified as different species according to their visible reproductive structures. *Penicillium dodgei* and *Eupenicillium brefeldianum* were classified as different species because they had different types of spores.

However, recently it was recognised that the spores of *P. dodgei* were asexual spores, while those of *E. brefeldianum* were sexual spores. A comparison of the DNA of these two fungi shows that they are the same species. This fungus is now known as *Penicillium brefeldianum*.

Outline how DNA analysis can show that *P. dodgei* and *E. brefeldianum* are the same species. [2]

- (d) Describe the advantages of using DNA analysis in determining homology between *P. dodgei* and *E. brefeldianum*. [3]

[Total: 12]

- 8 Fig. 8.1 shows part of the immune response to the first infection by a bacterial pathogen that has entered the body through the lining of a bronchiole. J and K are stages in the immune response.

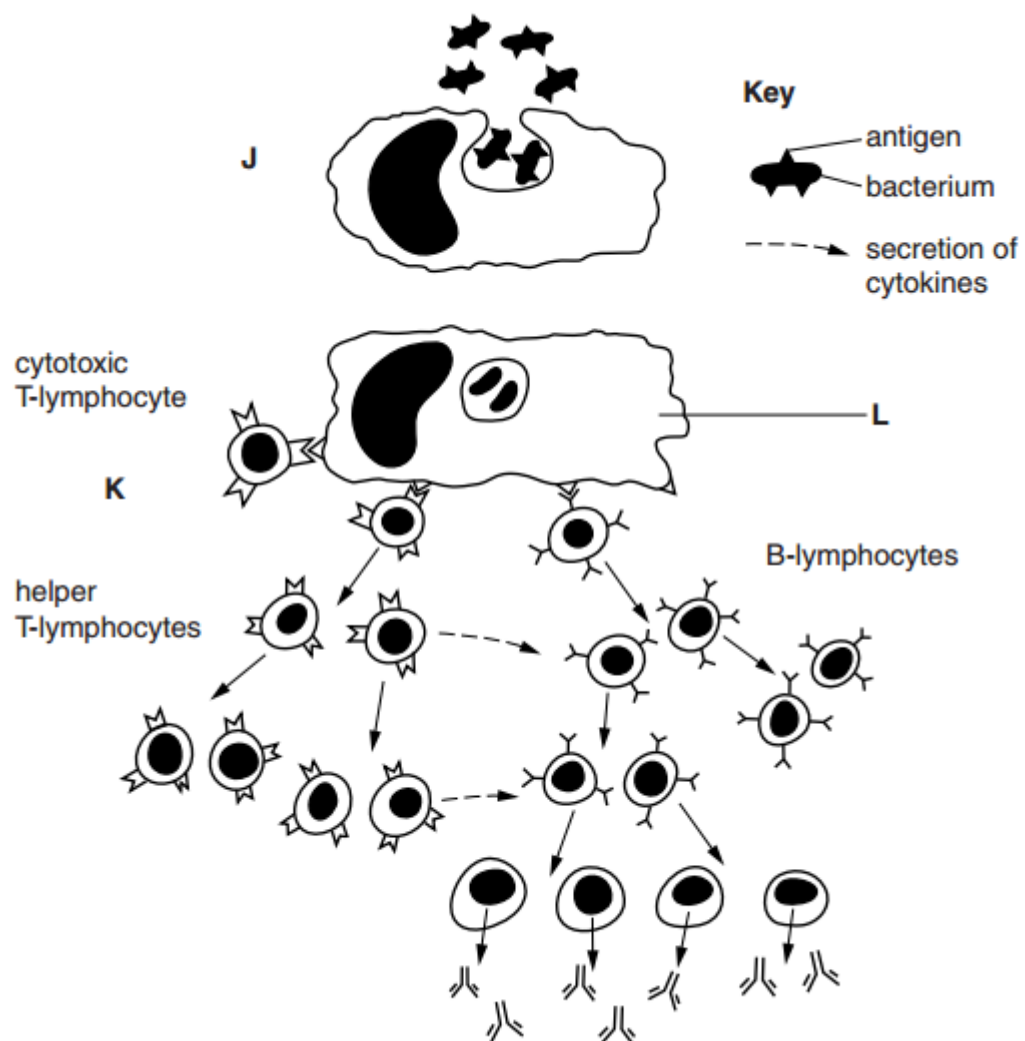


Fig. 8.1

(a) (i) State the process happening at stage J. [1]

(ii) Explain the role of cell L at stage K in the immune response. [2]

- (b) With reference to Fig. 8.1, explain how the response to a second infection by this bacterial pathogen differs from the first. [3]

B-lymphocytes have antibodies located on their external surface. When B-lymphocytes become plasma cells they then secrete antibodies.

Fig. 8.2 shows how the enzyme papain digests an antibody to obtain three fragments.

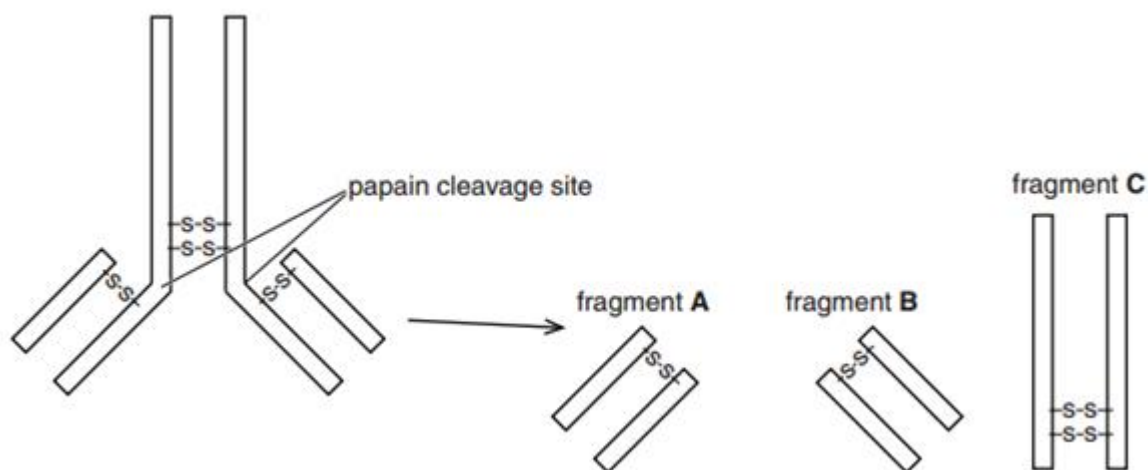


Fig. 8.2

- (c) The three fragments, A, B and C still retain their ability to function.

State the function of:

- (i) fragments A and B [1]

- (ii) fragment C. [1]

- (d)** There are various ways in which the effectiveness of immune responses can be reduced.

Suggest how each of the following reduces the effectiveness of an immune response.

- (i)** Some pathogens are covered in cell surface membranes from their host. [1]

- (ii)** B-lymphocytes do not mature properly and do not recognise any antigens. [1]

[Total: 10]

- 9** Reef-building corals are marine invertebrates closely related to jellyfishes and are found in shallow, clear tropical seas. The corals secrete an exoskeleton of calcium carbonate that becomes the underlying structure of the coral reef.

Zooxanthellae are a group of unicellular photosynthetic algae that live inside the cells of reef-building corals. The relationship is beneficial to both the zooxanthellae and the coral.

- (a)** Evidence shows that the relationship between zooxanthellae and reef-building corals has evolved by free-living algae invading corals that did not contain algae. [1]

- (i)** Corals that do not need zooxanthellae can live at a greater depth than reef-building corals.
Explain why. [3]

- (ii)** Suggest how the zooxanthellae may benefit in two ways from their association with the corals. [2]

Under conditions of environmental stress, the relationship between the reef-building corals and zooxanthellae can break down. Loss of zooxanthellae and the subsequent whitening that occurs, as shown in Fig. 9.1, is known as coral bleaching. Coral bleaching can lead to the death of the coral.



Fig. 9.1

- (b)** State one reason why permanent loss of zooxanthellae can lead to death of the coral. [1]

- (c)** One type of environmental stress that can cause coral bleaching is an increase in sea temperature.

Suggest why areas of sea with reef-building corals are particularly susceptible to increased temperature as a result of global climate change. [2]

[Total: 8]

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